

VibeSensor Diagnostic Report

Run date	Car	Duration	Sensors	Speed Band
2026-03-31 08:29:34 UTC	Unnamed Car — Sedan	02:25.4	4	80-90 km/h

Suspected source

Wheel / Tire

Inspect first

Front-Left

Action status

Action-ready with caution

Why this is first

Wheel / Tire remains the strongest source because the repeated pattern matched that source and stayed strongest near Front-Left in the 80-90 km/h window.

Why this corner wins

Dominant corner

Front-Left

Location confidence

Strong

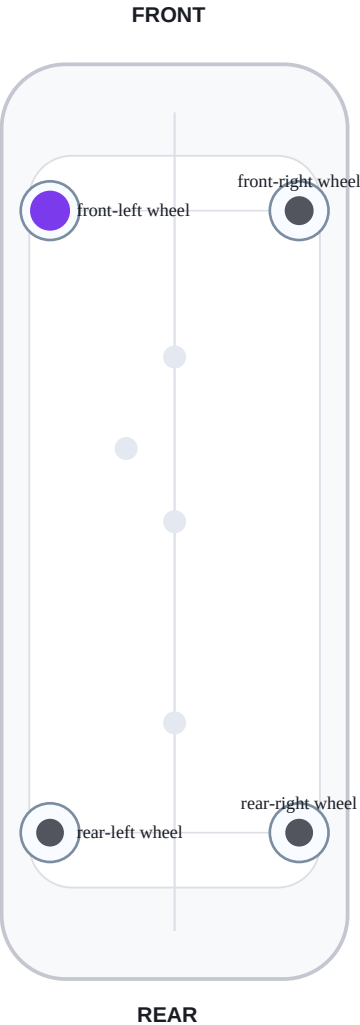
Coverage

4 of 4 wheel positions seen throughout run

Also consider

Driveline

Front-Left outranked the next location by 3.26x on matched-window linear intensity evidence.



What to do next

1

Check Front Left for tire damage, flat spots, belt shift, uneven wear, or pressure

Why: Tire structural issues often create strong 1x/2x wheel-order vibration.
Confirm: Visible/measurable tire defect aligns with complaint speed band.

2

Check Front Left for imbalance or radial/lateral runout.

Why: Wheel-order signatures are most likely caused by imbalance, runout, or tire deformation.
Confirm: A clear imbalance or runout is found and corrected, with vibration complaint reduced.

3

Check driveline mounts and couplings near Front Left for looseness or cracked rubber.

Why: Loose or degraded mounts can amplify normal order content into cabin vibration.
Confirm: Loose mount/fastener or cracked rubber support is found.

Need the inspection sequence? See A.

Need the evidence rows? See C.

Need the location proof? See B.

Need run details? See C.

Appendix A: Technician Worksheet

Primary vs alternative source

Primary source

Wheel / Tire

Alternative source

Driveline

Why the primary source is first

2x wheel, 1x wheel stayed strongest near Front-Left in the 80-90 km/h window.

If the primary path is clean

Move to the Driveline path next and inspect Front-Left.

Ranked source stack

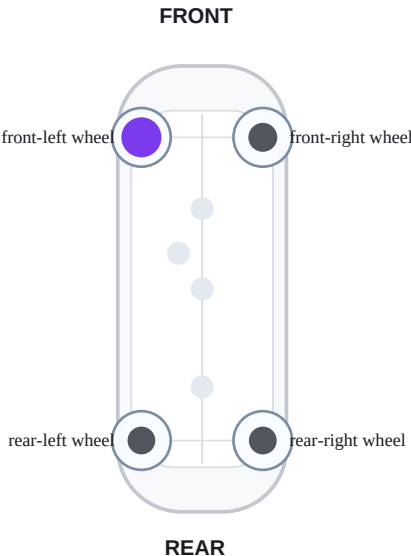
Source	Inspect first	Path role	Reason
Wheel / Tire	Front-Left	1. Primary path	2x wheel, 1x wheel stayed strongest near Front-Left in the 80-90 km/h window.
Driveline	Front-Left	2. Alternative path	1x driveshaft stayed strongest near Front-Left in the 80-90 km/h window.

Action matrix

Action	Why	Confirm	Falsify
Check Front Left for tire damage, flat spots, belt shift, uneven wear, or pressure mismatch.	Tire structural issues often create strong 1x/2x wheel-order vibration.	Visible/measurable tire defect aligns with complaint speed band.	No tire condition anomaly is found on inspected wheels.
Check Front Left for imbalance or radial/lateral runout.	Wheel-order signatures are most likely caused by imbalance, runout, or tire deformation.	A clear imbalance or runout is found and corrected, with vibration complaint reduced.	Balance and runout are within spec on all checked wheels/tires and complaint remains unchanged.
Check driveline mounts and couplings near Front Left for looseness or cracked rubber.	Loose or degraded mounts can amplify normal order content into cabin vibration.	Loose mount/fastener or cracked rubber support is found.	All inspected mounts and fasteners are within condition/torque spec.

Appendix B: Sensor Topology

Sensor map



Dominance summary

Dominant corner
Front-Left

Next strongest corner
Front-Right

Dominance ratio
3.26x matched-window linear intensity (not raw dB delta)

Computed from matched-window evidence; compare separately from the p95 dB snapshot below.

Location confidence
Strong

Coverage
4 of 4 wheel positions seen throughout run.

All expected positions remained connected throughout the run.

Location snapshot (context p95 dB)

Location	p95 dB	Coverage state
Front Left	36.6 dB	Complete
Front Right	34.6 dB	Complete
Rear Right	34.2 dB	Complete
Rear Left	34.1 dB	Complete

Appendix C: Evidence Detail

Evidence chain

Source	Signal	Measurement refs	Matched windows	Speed window	Location	Ambiguity
Wheel / Tire	2x wheel, 1x wheel	M01, M02, M03, M04	1894	80-90 km/h	Front-Left	—
Driveline	1x driveshaft	—	1894	80-90 km/h	Front-Left	Still in scope; no representative measurement appears below.

Supporting measurements

Measurement	Source	Signal	Peak dB	Strength dB	Speed window	Location
M01	Wheel / Tire	2x wheel, 1x wheel	43.9 dB	43.8 dB	80-90 km/h	Front-Left
M02	Wheel / Tire	2x wheel, 1x wheel	34.3 dB	33.9 dB	80-90 km/h	Front-Left
M03	Wheel / Tire	2x wheel, 1x wheel	29.5 dB	28.8 dB	80-90 km/h	Front-Left
M04	Wheel / Tire	2x wheel, 1x wheel	25.1 dB	24.3 dB	80-90 km/h	Front-Left

Supporting context

Speed-band summary
80-90 km/h

Phase summary
Cruise

Suitability detail

Speed variation: Speed range stayed in a usable diagnostic band for steady-state diagnosis and order tracking.
Sensor coverage: Multiple sensor locations observed.
Reference completeness: Required order references are present.
Saturation and outliers: No obvious saturation detected.
Frame integrity: No dropped frames or queue overflows detected.

Traceability

Appendix C: Run Context
Run ID
6fabed5cba7e48b9aa38d38c7829d23f
Run date
2026-03-31 08:29:34 UTC
Start Time (UTC)
2026-03-31 08:27:08 UTC
End Time (UTC)
2026-03-31 08:29:34 UTC
Car
Unnamed Car — Sedan
Sensors
front_left, front_right, rear_left, rear_right
Sensor Model
ADXL345
Firmware Version
None
Sample count
1934
Raw Sample Rate (Hz)
800
tire size
285/30R21
Export package
6fabed5cba7e48b9aa38d38c7829d23f_export.zip
Report version
v0.0.0-dev